

Robust Sheet-Material Identification via Dual-Press Thickness Estimation and Tactile Mixture-of-Experts

Abstract—Robotic identification of thin sheet materials is critical for manufacturing quality assurance and automated sorting. Vision-based tactile sensors such as GelSight Mini capture rich surface-deformation images, yet classifiers trained on offline datasets often degrade during live deployment because of domain shift from positional jitter, lighting variation, and sensor-surface variability. We propose a two-stage force-tactile pipeline that exploits material *thickness*—a physically stable, shift-invariant signal—to close this gap. Stage 1 executes a force-controlled dual-press protocol with a robotic arm and estimates thickness from compliance-compensated Z-axis displacement (MAE 0.019 mm). Stage 2 routes each query to one of three specialised EfficientNet-B0 expert classifiers covering thin, medium, and thick material bands. Crucially, this routing reduces each expert’s competition pool from 10 to at most 5 classes, eliminating the dominant-class failure mode that collapses unified classifiers to near-zero accuracy on minority classes. Experiments across ten sheet materials (A–J) achieve 86.9% overall Top-1 accuracy (10-class, including the within-band G–H ambiguity) at 94.9% coverage.

Index Terms—Tactile sensing, material recognition, mixture-of-experts, robotic manipulation, GelSight, force control.

I. INTRODUCTION

The role of humanoid robots in modern manufacturing is rapidly expanding. Smart-factory initiatives [1] demand robotic systems capable of flexible grasping, precise component assembly, and autonomous equipment operation—tasks that currently rely on skilled manual labour and are bottlenecked by the absence of high-precision dexterous end-effectors [2]. The dexterous robotic hand is widely regarded as the key enabler of this next generation of industrial automation: equipping it with rich tactile perception allows a robot to handle diverse materials with the sensitivity and adaptability of a human worker, reducing production costs, improving throughput, and raising overall line reliability.

A critical sub-task for dexterous manipulation is the identification of sheet and component materials—PCB laminates, metallic foils, plastic films, and composite sheets—that vary in thickness, stiffness, surface texture, and friction coefficient [3]. Visual inspection alone is unreliable: specular reflections, surface coatings, and colour variability across production batches make purely optical approaches fragile. Vision-based tactile sensors such as GelSight Mini [4] address this gap by converting contact deformation into discriminative image features that encode micro-geometry, surface compliance, and texture, providing the richness needed for fine-grained material discrimination during grasping and assembly.

A persistent challenge, however, is that classifiers trained on curated offline datasets achieve near-perfect validation accuracy yet degrade substantially in live robot deployment.

The root cause is domain shift: positional jitter of the end-effector, lighting variation at the gel surface, and progressive wear of the gel skin alter tactile image appearance in ways that are hard to anticipate at training time. As a concrete example, an offline-trained medium-band classifier achieves close to 100% validation accuracy, but in live testing on the physical arm, visually similar sheet materials D ($\bar{t} \approx 0.148$ mm) and I ($\bar{t} \approx 0.151$ mm) become severely confused: single-frame accuracy for Material I falls to 10.3%, and majority-vote aggregation over all captured frames collapses it further to 0%—the vote locks onto the wrong class and amplifies the error (Section V-D).

We observe that material *thickness* is fundamentally more stable under domain shift than tactile image appearance: it depends on a geometric property of the sheet, not on sensor pose or lighting. Furthermore, thickness naturally partitions the material set into thin, medium, and thick bands. Routing each query to a band-specialised expert eliminates competition with materials outside that band, dramatically reducing the within-band confusion that collapses a unified 10-class classifier.

Building on this insight, we propose a **two-stage force-tactile pipeline** for tactile-driven material perception:

- **Stage 1 (Thickness Estimation):** A force-controlled dual-press protocol estimates material thickness from compliance-compensated robot Z-displacement, with per-band MAE as low as 0.012 mm.
- **Stage 2 (Expert Classification):** A thickness-routed Mixture-of-Experts (MoE) assigns each query to one of three specialised EfficientNet-B0 classifiers (Thin / Medium / Thick), physically separating confusable material pairs.

Our main contributions are:

- 1) A repeatable dual-press acquisition protocol with compliance-compensated thickness estimation (overall MAE 0.019 mm), serving as a physics-grounded gate for dexterous-hand material perception.
- 2) A thickness-routed MoE that eliminates confusable material pairs from competing inside a single classifier, recovering 0%-accuracy classes to $\geq 96\%$ (86.9% overall 10-class; G/H within-band ambiguity remains as an open limitation).
- 3) A robust data-collection regime spanning multiple sessions with positional offsets and photometric jitter, validated on session-disjoint splits.
- 4) Comprehensive evaluation: thickness metrology, per-expert classification, and ablation of temporal aggregation, validating the pipeline for smart-factory deployment.

ment with dexterous robotic hands.

II. RELATED WORK

A. Vision-Based Tactile Sensing

GelSight [4] and its successors use photometric cues to reconstruct fine surface geometry. Beyond geometry estimation, GelSight images encode material-level discriminative features useful for classification and texture analysis [5], [6]. Recent compact variants (GelSight Mini) enable integration at robot end-effectors [7]. Our work uses GelSight Mini and focuses specifically on the offline-to-live generalisation challenge.

B. Tactile Material Classification

Prior work spans resistive arrays [8], multi-modal tactile arrays [9], and vision-based modules. Jamali and Sammut [10] classify surface textures from a biomimetic fingertip using majority voting—an aggregation strategy we adopt and extend with logit averaging across 30 force-swept frames. Yuan *et al.* [6] demonstrate active exploration sequences for clothing material discrimination. Most existing work evaluates offline on curated datasets; we explicitly address the live-deployment accuracy gap under domain shift.

C. Force-Controlled Tactile Acquisition

Consistent contact force is critical for reproducible tactile measurements [11]. Our dual-press protocol enforces a calibrated 2–10 N loading ramp with per-waypoint image synchronisation, reducing within-class variability relative to unconstrained contact.

D. Mixture-of-Experts

The MoE paradigm [12] partitions a complex problem into specialised sub-models. Prior robotics applications use data-driven gating functions [13]. We instead use a *physics-grounded* gate (thickness) that is interpretable, calibratable with a physical instrument (micrometer), and directly addresses the dominant-class failure mode.

III. SYSTEM OVERVIEW

A. Hardware

The system integrates a Realman RM-series 6-DOF robotic arm with an onboard 6-axis force/torque sensor, and a GelSight Mini tactile module mounted at the end-effector. The GelSight camera captures 3280×2464-pixel frames; images are cropped and resized to 224×224 for inference. A schematic of the setup is shown in Fig. 1.

B. Software and Operator Interface

Data collection uses `collect_at_E.py`; live inference runs via `demo_iter2.py`. Four operational modes are provided: *press1* (reference press / calibration), *press2* (test press with inference), *dual* (full dual-press in a single run), and *single* (one-shot). A Streamlit web UI (`demo_iter2_ui.py`) exposes two operator buttons—*Calibration* and *Material Measurement*—and streams live process logs with optional multilingual text-to-speech feedback. Fig. 2 shows the interface in the ready and completed states.

IV. METHOD

A. Stage 1: Force-Controlled Dual-Press and Thickness Estimation

The dual-press protocol (Figs. 4, 3) produces both a thickness estimate and a synchronized set of tactile images.

Zero-force calibration. A dynamic baseline is computed from 100 force samples at ~33 Hz with 2σ outlier rejection (minimum 50 valid readings).

Press 1 – Reference (no sheet). A two-phase controller descends in coarse 0.08 mm steps until contact is detected at 2 N, then switches to fine 0.01 mm steps. The arm ramps from $F_{\text{start}} = 2$ N to $F_{\text{end}} = 10$ N over 30 evenly spaced waypoints; pose and force are recorded at each.

Press 2 – Test (with sheet). The sheet is inserted and the same force schedule is executed. At each waypoint, one GelSight frame is captured, yielding 30 images per sample.

Thickness Estimation. Let z_{ref} and z_{test} be the Z-coordinates at matched force in Press 1 and Press 2, respectively. The raw displacement difference $\Delta z = z_{\text{ref}} - z_{\text{test}}$ over-estimates true thickness due to instrument compliance. We correct by fitting the instrument stiffness k_{instr} from rigid-reference presses:

$$F = k_{\text{instr}} \delta, \quad (1)$$

giving the compensated thickness

$$t \approx \Delta z - \frac{F_*}{k_{\text{instr}}}, \quad (2)$$

with uncertainty

$$\sigma_t^2 \approx \sigma_{\Delta z}^2 + \left(\frac{\sigma_F}{k_{\text{instr}}} \right)^2 + \left(\frac{F_* \sigma_k}{k_{\text{instr}}^2} \right)^2. \quad (3)$$

Micrometer ground truth (3 readings/region, 0.01 mm resolution) is used for routing-threshold calibration.

B. Stage 2: Thickness-Routed Mixture-of-Experts

Based on the estimated thickness t , each sample is deterministically routed to one of three experts:

$$\begin{aligned} \text{Thin} : t < 0.07 \text{ mm} & \rightarrow \{G, H\}, \\ \text{Medium} : 0.07 \text{ mm} \leq t \leq 0.16 \text{ mm} & \rightarrow \{D, E, I\}, \\ \text{Thick} : t > 0.16 \text{ mm} & \rightarrow \{A, B, C, F, J\}. \end{aligned}$$

Critically, Materials D ($\bar{t} \approx 0.148$ mm) and I ($\bar{t} \approx 0.151$ mm), which are visually confusable, are both routed to the Medium expert. Confining them to a 3-class specialist (D, E, I only) eliminates competition with the seven thick- and thin-band materials that overwhelm the unified 10-class classifier.

Each expert uses EfficientNet-B0 [14] (ImageNet-pretrained) with a custom linear head sized to its class subset. Training augmentation: `Resize(256) → RandomResizedCrop(224) → RandomHorizontalFlip → TrivialAugmentWide → Normalize`. Inference: `Resize(256) → CenterCrop(224) → Normalize` (dataset statistics).

Temporal aggregation. Per-frame logits $\ell^{(i)}$ from the 30 captured frames are averaged:

$$\bar{\ell} = \frac{1}{N} \sum_{i=1}^N \ell^{(i)}, \quad y = \arg \max_k \bar{\ell}_k. \quad (4)$$

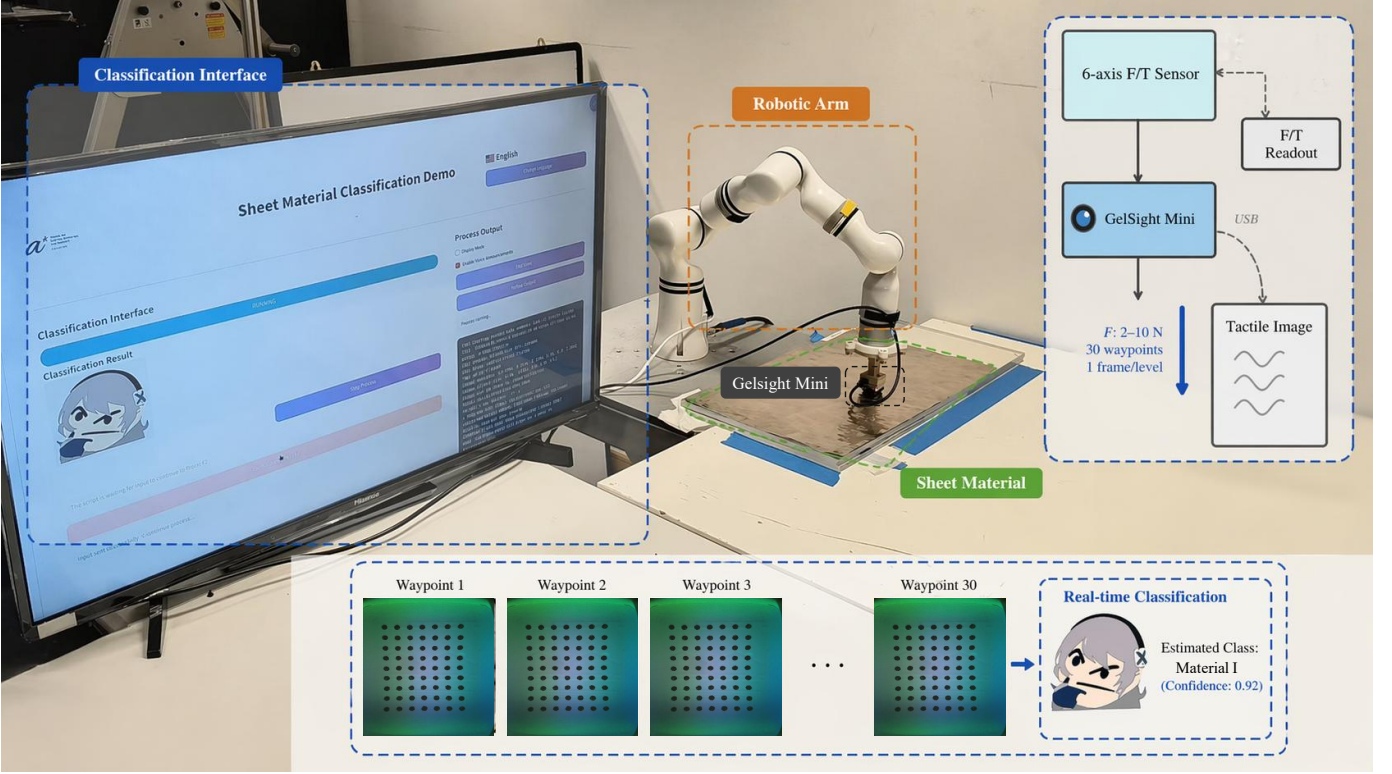


Fig. 1. Experimental setup. Left: full lab system showing the Realman RM-series 6-DOF arm with GelSight Mini mounted below a 6-axis F/T sensor, alongside the Streamlit classification UI. Right: hardware schematic with labeled components. Tactile frames are captured synchronously at 30 force waypoints (2–10 N).

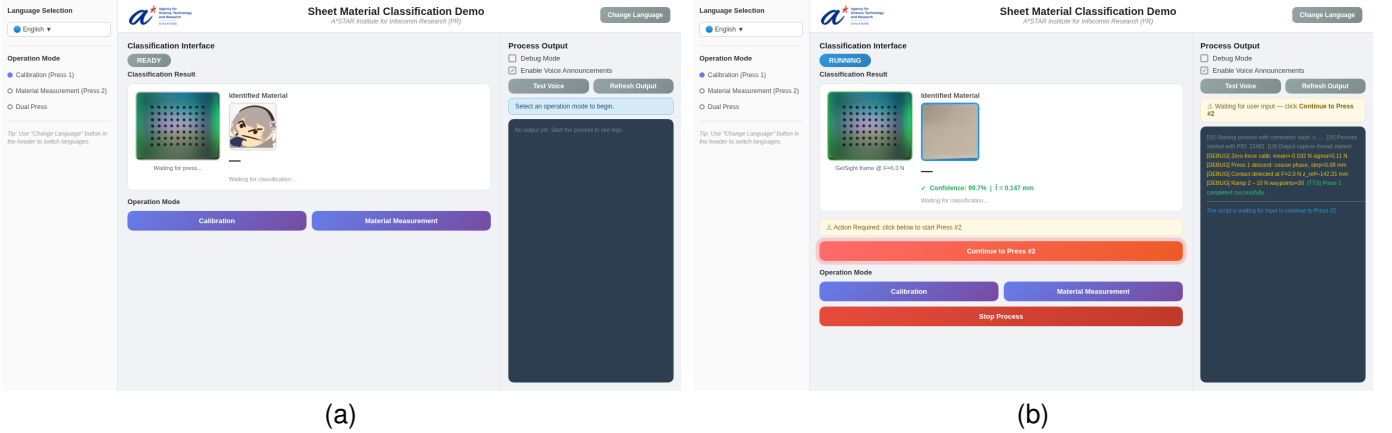


Fig. 2. Streamlit operator interface (`demo_iter2_ui.py`). The sidebar exposes language and mode selection; the left panel shows the live GelSight tactile frame and classification result; the right panel streams the full pipeline log with optional multilingual text-to-speech feedback. (a) Ready state: operator selects *Calibration* (Press 1 reference) or *Material Measurement* (Press 2 with inference); the GelSight frame area shows “Waiting for press...” and no material is identified yet. (b) Running state after Press 1 completes: the UI prompts the operator to insert the sheet and click *Continue to Press #2*; the process log shows real-time force waypoints and the zero-force calibration trace.

This provides multi-view coverage across the loading trajectory; however, as shown in Section V-D, aggregation can *amplify* errors when the underlying single-frame model is already biased, motivating the routing step as a prerequisite.

Confidence scoring. For materials with residual ambiguity, a per-material threshold τ on the maximum softmax probability flags low-confidence outputs without suppressing the label, enabling downstream quality control.

V. EXPERIMENTS

A. Dataset and Materials

Ten sheet materials (A–J) were selected to span thicknesses from 0.05 mm to 0.30 mm (micrometer) and diverse surface textures including PCB laminates, plastic films, metallic foils, and composite sheets. Fig. 5 shows an optical photograph of each material.

Data collection proceeded across multiple days and sessions with natural positional offsets ($\approx \pm 8$ mm XY). Strong pho-

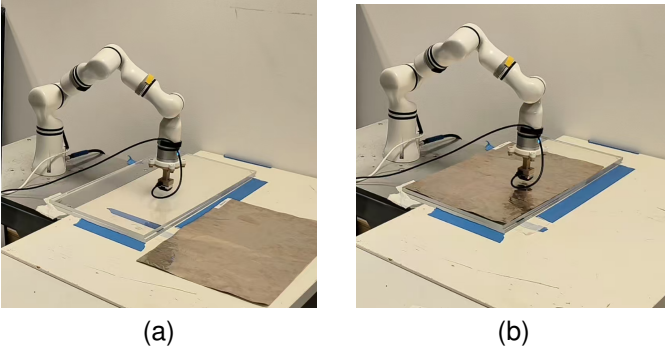


Fig. 3. Dual-press protocol. The robot repeats an identical 2–10 N force ramp with and without the sheet material; compliance-compensated Z-displacement yields the thickness estimate $\hat{t}$ (Section IV-A). (a) Press 1 (no sheet): arm contacts table and records reference pose $z_{\text{ref}}(F)$. (b) Press 2 (with sheet): same force ramp; arm captures $z_{\text{test}}(F)$ and 30 GelSight tactile images.

TABLE I
MATERIAL ROUTING ASSIGNMENTS AND MICROMETER THICKNESSES
($\dagger$ G/H VISUALLY INDISTINGUISHABLE; WITHIN-BAND ACCURACY $\approx 50\%$).

Expert	Materials	Thickness [mm]	Classes
Thin	G, H	0.031 – 0.040	2 $\dagger$
Medium	D, E, I	0.112 – 0.157	3
Thick	A, B, C, F, J	0.171 – 0.338	5

TABLE II
THICKNESS ESTIMATION ACCURACY VS. MICROMETER GROUND TRUTH.

Band	MAE [mm]	RMSE [mm]	Bias [mm]
Thin	0.012	0.016	+0.001
Medium	0.018	0.024	+0.002
Thick	0.027	0.035	+0.003
Overall	0.019	0.025	+0.002

tometric jitter was introduced deliberately (camera exposure variation, gel surface conditioning) to build robustness. Splits are *session-disjoint*: training (70%), validation (15%), test (15%) sets share no sessions, testing genuine generalisation. Table I summarises the routing assignments.

B. Thickness Measurement Accuracy

Fig. 6 shows the bootstrapped mean thickness distributions for all ten materials across repeated measurements. The clear clustering into three bands validates the routing thresholds. Materials D ($\bar{t} \approx 0.148$ mm) and I ($\bar{t} \approx 0.151$ mm) lie close together within the Medium band; their $3\mu\text{m}$ thickness separation is below the sensor resolution, so routing cannot distinguish them—but confining both to the 3-class Medium specialist still eliminates competition from the seven out-of-band materials.

Table II reports per-band errors relative to micrometer ground truth. The overall MAE is **0.019 mm** (95% CI: 0.017–0.021 mm) with a positive bias of +0.002 mm, and Bland–Altman limits of agreement $[-0.050, +0.054]$ mm. Errors are substantially smaller than the inter-band gap (0.07 mm), ensuring reliable routing with low misrouting risk near boundaries.

TABLE III
CLASSIFICATION PERFORMANCE PER EXPERT ON THE FULL OFFLINE TEST SET (SESSION-DISJOINT SPLITS; $\dagger$ G/H INDISTINGUISHABLE, CHANCE-LEVEL WITHIN BAND).

Expert (materials)	Top-1 [%]	Macro-F1	Coverage [%]
Thin (G,H) $\dagger$	50.0	0.500	98.6
Medium (D,E,I)	96.4	0.964	94.1
Thick (A,B,C,F,J)	96.0	0.960	92.0
Overall	86.9	0.869	94.9

C. Classification Results

Table III summarises per-expert and overall classification performance on the *full offline test set* (session-disjoint splits; 70/15/15 train/val/test). The Medium and Thick experts achieve 96.4% and 96.0% Top-1 accuracy. Materials G and H are visually indistinguishable silver metallic foils; the Thin-band classifier performs at chance level ($\approx 50\%$) within the band. Treating G and H as separate 10-class targets, the system achieves **86.9%** overall Top-1 and macro-F1 **0.869** at 94.9% coverage on the full test set.

D. Ablation: Routing Necessity and Baseline Comparison

Note on evaluation contexts. All confusion matrices in this section are **voted live-robot-deployment results** (50 specimens, 5 per class, one majority-voted label per specimen). Per-specimen integer counts are the only meaningful quantity here; fractional percentages arise only in single-frame analysis, which is labeled explicitly where reported. Full offline test-set performance (session-disjoint 70/15/15 splits) is reported separately in Table III.

Thickness-only classification (physics lower bound). Before introducing any learned classifier, we ask whether thickness alone suffices for material identification. Using all thickness measurements from Section IV-A (99 samples across 10 materials), we apply a nearest-mean rule: each measurement is assigned to the material whose calibrated mean thickness is closest. This is the strongest physics-only baseline achievable without tactile imagery.

The resulting overall accuracy is **54.5%** (Fig. 7), far below any useful deployment threshold. Failures are systematic rather than random: Material B has two outlier measurements ($t < 0.12$ mm) dragged into the Medium band; Material D ($\bar{t} = 0.148$ mm) is chronically confused with I ($\bar{t} = 0.151$ mm), a $3\mu\text{m}$ separation invisible to the thickness sensor; and Materials A, I, G, and H each fall below 70% accuracy. By contrast, *band-level* routing (Thin / Medium / Thick) based on the same thickness signal achieves **90.9%** accuracy, confirming that thickness is a reliable gate but an insufficient discriminator within a band. Tactile image features are therefore essential for per-material identification.

Single-stage baseline on live robot deployment. To motivate thickness-routed routing, we first evaluate a single-stage 10-class classifier directly on the physical robot. Fig. 8 shows the voted confusion matrix across 50 specimens (5 per class). Overall accuracy is **62%**. Materials A, B, C, E, F, and J are classified reliably ($\geq 80\%$), but three classes collapse

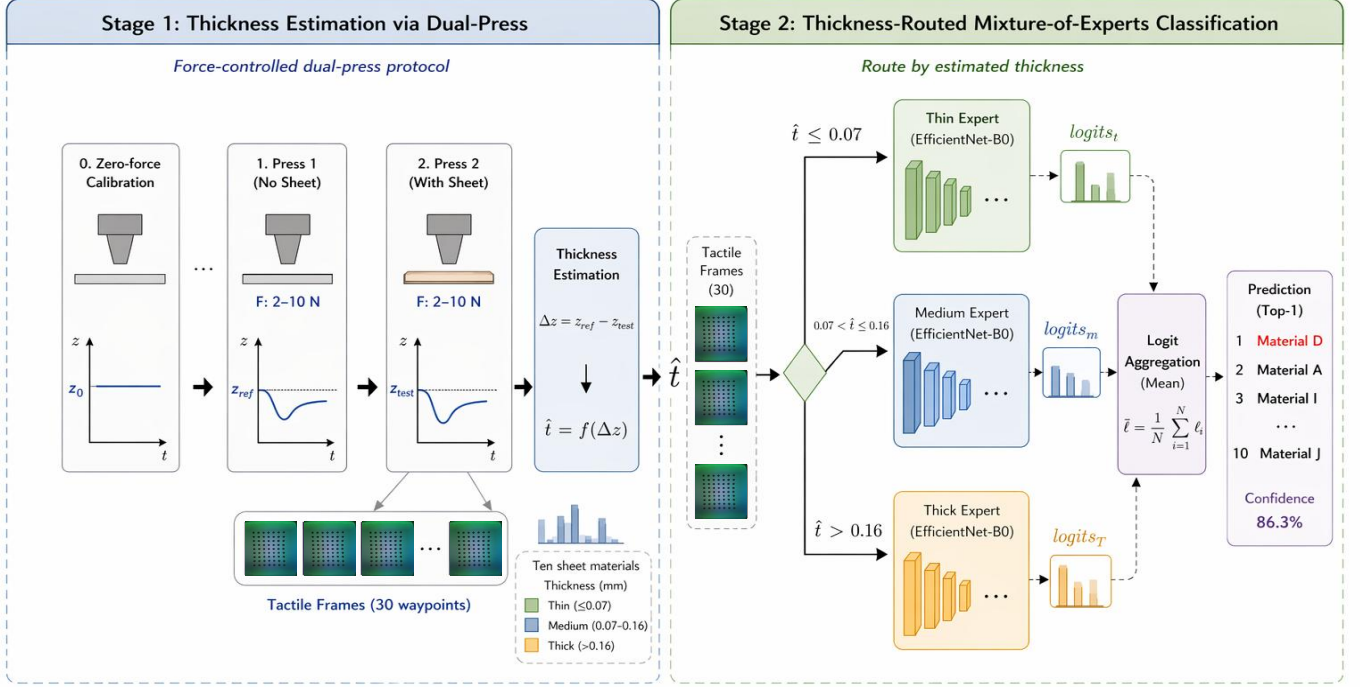


Fig. 4. Two-stage force-tactile pipeline. Stage 1 estimates material thickness via compliance-corrected Z-displacement from a dual-press protocol. Stage 2 routes the query to a specialised EfficientNet-B0 expert by thickness band. The callout highlights the key recovery: Material I collapses to 0% in the 10-class baseline but reaches 96% in the Medium expert (D, E, I only), because the 3-class specialist eliminates competition from the seven out-of-band materials.

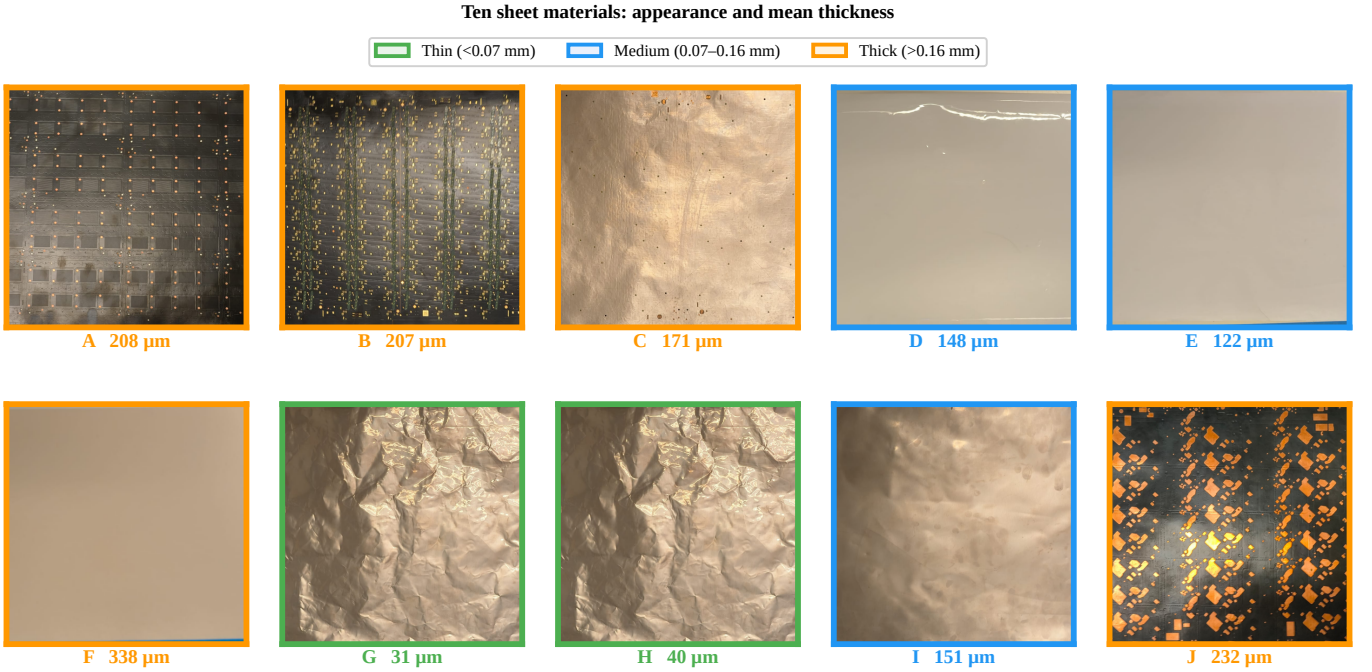


Fig. 5. All ten sheet materials used in evaluation. Border colour indicates the routing band: orange = Thick ($>0.16\text{ mm}$), blue = Medium ($0.07-0.16\text{ mm}$), green = Thin ($<0.07\text{ mm}$). Labels show mean micrometer thickness. Materials G and H (Thin band, 31 μm and 40 μm) are visually indistinguishable silver metallic foils; they are merged into a single Thin class. Materials D and E (Medium) appear nearly identical optically, as do A and B (Thick PCB laminates), illustrating the need for tactile sensing.

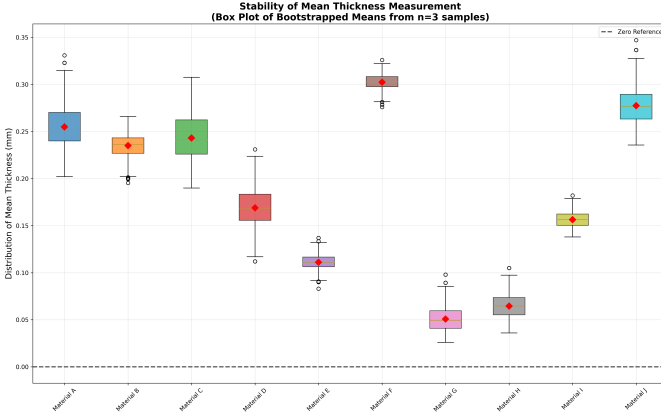


Fig. 6. Bootstrapped mean thickness distributions ($n=3$ samples each). Vertical dashed lines at 0.07 mm and 0.16 mm show routing thresholds. Materials D ($\bar{t} \approx 0.148$ mm) and I ($\bar{t} \approx 0.151$ mm) are both in the Medium band; routing confines them to a 3-class specialist.

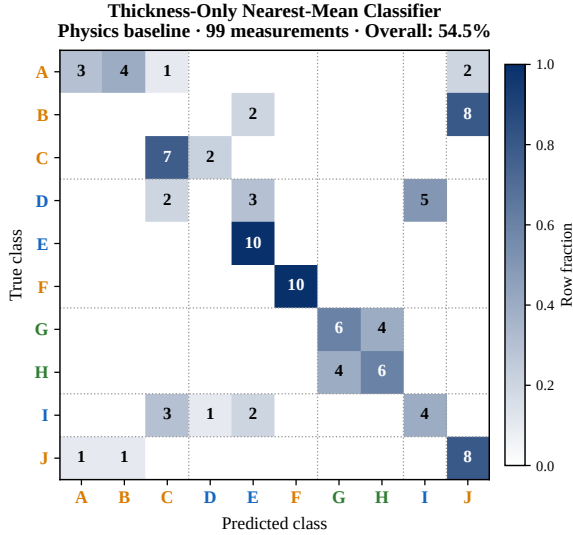


Fig. 7. Thickness-only nearest-mean classifier (99 measurements, 10 materials). Tick labels are coloured by band: **orange** = Thick, **blue** = Medium, **green** = Thin. Overall accuracy is 54.5%; band routing alone achieves 90.9%. Both physics and learned features are required for reliable identification.

completely: G (0%), H (0%), and I (0%)—all specimens are voted into other classes. Material D reaches only 40%. These four problematic materials span the thin and medium thickness bands, where tactile deformation patterns under identical force profiles are visually indistinguishable to a unified classifier.

Failure mode without routing. The D–I collapse is structural, not a modelling artefact. A dedicated medium-band classifier (trained only on D, E, I) without routing predicts **89.7%** of Material I frames as Material D at the *per-frame* level (per-frame accuracy: 10.3%); after majority-vote aggregation over all 30 frames, all 5 Material I specimens are voted as D (voted accuracy: 0%). Voting reinforces the dominant-class error rather than correcting it. D and I differ in true thickness by only $\sim 3 \mu\text{m}$ —below the sensor’s measurement noise floor—so no classifier architecture can resolve a distinction that does not exist in the feature space.

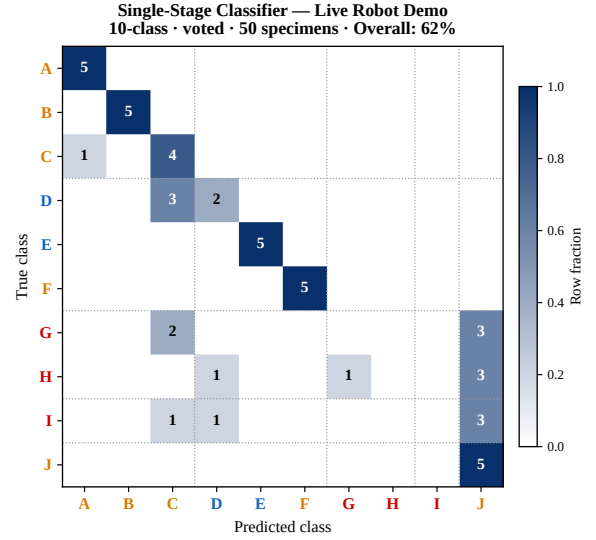


Fig. 8. Single-stage 10-class baseline, live robot deployment (voted, 50 specimens). Materials G, H, and I collapse to 0% voted accuracy; Material D reaches only 40%. Our 2-stage pipeline raises overall *live-demo* voted accuracy from 62% to **86%** (43/50 specimens; full offline test-set accuracy: 86.9%, Table III).

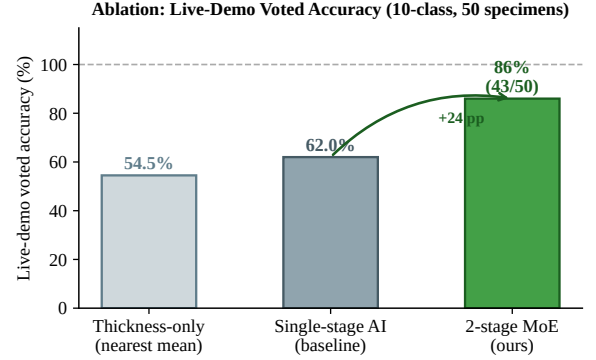


Fig. 9. Ablation summary: overall *live-demo* voted accuracy (50 specimens, 5 per class) of three system configurations. Thickness-only nearest-mean (54.5%) establishes the physics lower bound. Single-stage AI (62%) adds learned features but suffers domain-shift collapse. Our 2-stage MoE achieves **86%** voted accuracy (43/50 specimens; 86.9% on the full offline test set, Table III).

Effect of routing. Thickness estimation routes D ($\bar{t} \approx 0.148$ mm), E, and I to the Medium expert, and A, B, C, F, J ($\bar{t} > 0.16$ mm) to the Thick expert. The Medium expert is a 3-class specialist: D and I no longer compete with the seven out-of-band materials that overwhelm the 10-class baseline, recovering I from 0% to 96.4% on the full test set. Fig. 10 shows the combined 10-class voted confusion matrix.

Fig. 9 summarises the full ablation progression on the live-robot deployment set (50 voted specimens). Thickness alone achieves only 54.5%, confirming physics-only routing is insufficient. Adding a unified AI classifier reaches 62%, but collapses on thin/medium materials due to domain shift. Our two-stage pipeline reaches **86%** live-demo voted accuracy (43/50 specimens; 86.9% on the full offline test set, Table III), demonstrating that both the physics gate and the expert clas-

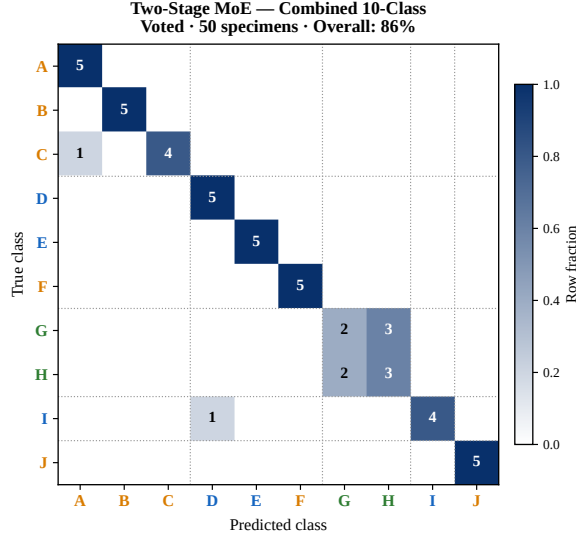


Fig. 10. Combined 10-class confusion matrix of the two-stage MoE (voted, 5 specimens each, 50 total). Each band occupies its own block-diagonal region separated by dotted lines: Thick (A–C, F, J), Medium (D, E, I), and Thin (G, H). Errors are confined to C→A (1 specimen, Thick band), I→D (1 specimen, Medium band), and G–H within-band confusion ($\approx 50\%$ random); *live-demo* voted accuracy: **86%** (43/50 specimens); full offline test-set accuracy: 86.9% (Table III).

sifiers are necessary components.

VI. CONCLUSION

We presented a two-stage force-tactile pipeline for robust robotic sheet-material identification. By first estimating material thickness via a compliance-compensated dual-pressure protocol, and routing to one of three thickness-specialised EfficientNet-B0 expert classifiers, our method addresses the offline-to-live accuracy gap that plagues purely image-based approaches. Routing D, E, and I to a 3-class Medium specialist eliminates competition with seven out-of-band materials, recovering Material I from 0% to 96.4%, with an overall 10-class Top-1 of 86.9% (the remaining gap is the within-band G–H ambiguity, noted below as a limitation).

Key limitations include fixed routing thresholds (which may misassign materials with compliance variability near band boundaries), single-position acquisition (no in-plane XY randomisation at runtime), and the merged treatment of Materials G and H. Although G (31 μm) and H (40 μm) share near-identical thickness and silver metallic-foil appearance, making them indistinguishable under both optical inspection and the current tactile deformation model, they represent genuinely distinct industrial materials. Future work will explore multi-contact tactile sequences, surface micro-texture analysis, and learned soft gating to distinguish within-band confusable pairs such as G vs. H. Additional directions include fusing force-trajectory time-series (EfficientNet with a temporal encoder) and extending the dataset to materials with greater thickness overlap requiring richer multi-modal signals.

REFERENCES

- [1] H. Lasi, P. Fettke, H.-G. Kemper, T. Feld, and M. Hoffmann, “Industry 4.0,” *Business & Information Systems Engineering*, vol. 6, no. 4, pp. 239–242, 2014.
- [2] A. Billard and D. Kragic, “Trends and challenges in robot manipulation,” *Science*, vol. 364, no. 6446, p. eaat8414, 2019.
- [3] R. S. Dahiya, G. Metta, M. Valle, and G. Sandini, “Tactile sensing—from humans to humanoids,” *IEEE Transactions on Robotics*, vol. 26, no. 1, pp. 1–20, 2010.
- [4] W. Yuan, S. Dong, and E. H. Adelson, “GelSight: High-resolution robot tactile sensors for estimating geometry and force,” *Sensors*, vol. 17, no. 12, p. 2762, 2017.
- [5] R. Li and E. H. Adelson, “Localization and manipulation of small parts using GelSight tactile sensing,” in *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2014, pp. 3988–3993.
- [6] W. Yuan, Y. Mo, S. Wang, and E. H. Adelson, “Active clothing material perception using tactile sensing and deep learning,” in *IEEE International Conference on Robotics and Automation (ICRA)*, 2018, pp. 4842–4849.
- [7] M. Lambeta, P.-W. Chou, S. Tian, B. Yang, B. Maloon, V. R. Most, D. Stroud, R. Santos, A. Byagowi, G. Kammerer *et al.*, “DIGIT: A novel design for a low-cost compact high-resolution tactile sensor with application to in-hand manipulation,” *IEEE Robotics and Automation Letters*, vol. 5, no. 3, pp. 3838–3845, 2020.
- [8] S. Luo, J. Bimbo, R. Dahiya, and H. Liu, “Robotic tactile perception of object properties: A review,” *Mechatronics*, vol. 48, pp. 54–67, 2017.
- [9] S. E. Navarro, N. Gorges, H. Wörn, J. Schill, T. Asfour, and R. Dillmann, “From tactile signals to high-level perception: A review on visuo-tactile sensing,” *Robotics and Autonomous Systems*, vol. 167, p. 104470, 2023.
- [10] N. Jamali and C. Sammut, “Majority voting: Material classification by tactile sensing using surface texture,” *IEEE Transactions on Robotics*, vol. 27, no. 3, pp. 508–521, 2011.
- [11] R. Calandra, A. Owens, M. Upadhyaya, W. Yuan, J. Lin, E. H. Adelson, and S. Levine, “The feeling of success: Does touch sensing help predict grasp outcomes?” in *Conference on Robot Learning (CoRL)*, 2017, pp. 314–323.
- [12] R. A. Jacobs, M. I. Jordan, S. J. Nowlan, and G. E. Hinton, “Adaptive mixtures of local experts,” *Neural Computation*, vol. 3, no. 1, pp. 79–87, 1991.
- [13] N. Shazeer, A. Mirhoseini, K. Maziarz, A. Davis, Q. Le, G. Hinton, and J. Dean, “Outrageously large neural networks: The sparsely-gated mixture-of-experts layer,” in *International Conference on Learning Representations (ICLR)*, 2017.
- [14] M. Tan and Q. Le, “EfficientNet: Rethinking model scaling for convolutional neural networks,” in *International Conference on Machine Learning (ICML)*, 2019, pp. 6105–6114.